

Cumulative Reflection

During my time in college as a Computer Engineering student, I learned that engineering is not only about writing code or building circuits. It is also about communicating clearly, solving problems step by step, and making responsible decisions that affect people and society. In this reflection, I will explain how I improved my communication skills, how I learned to design and solve engineering problems, how I used outside resources to learn, and how I understand the impact of engineering in society. I will also explain how I plan to continue learning in the future. One of the biggest areas where I improved is communication. In many of my projects, I worked in teams. At first, I focused only on finishing my part of the work. Later, I realized that good communication is very important. When team members clearly explain their progress, problems can be solved faster. I learned to ask clear questions and give regular updates. I also learned that communication is not only about speaking. Writing clearly is just as important. Writing project summaries, organizing information, and explaining results in a simple way helps others understand the work quickly. Over time, my writing became more organized, with clear goals, explanations, and conclusions.

My classes helped me build a strong foundation in engineering. In cpre 2810 Digital Logic Design, I learned how digital circuits work at a basic level. I learned about logic gates such as AND, OR, and NOT, and how they connect together to form larger circuits. I studied adders, multiplexers, decoders, and registers. This class helped me understand how computers are built from small building blocks. I realized that complex systems are created by combining simple parts carefully. Understanding the basics made advanced topics easier later. Another important class was cpre 2880 Embedded Systems. In this course, I learned how to program hardware and control real devices. One exciting project was working with the Roomba robot. I learned how sensors detect information and how motors move the robot. I also learned about timers, interrupts, and input/output pins. This class taught me that timing is very important in embedded systems. A system must respond at the correct time, or it will not work properly. I also improved my debugging skills. When the robot did not behave correctly, I had to check the code, sensor values, and control logic step by step to find the problem. My favorite class was cpre 3810 computer architecture and organization. In this class, I designed a RISC-V processor. This project helped me understand how a CPU works internally. I learned how instructions are fetched, decoded, and executed. I studied how data moves through registers and arithmetic units, and how the control unit sends signals. Designing the processor taught me how to break a large system into smaller parts and connect them correctly. Testing the processor with simulations was very rewarding. When it successfully ran instructions, I felt proud of my work. This experience increased my interest in hardware and chip design.

Another important skill I developed is problem-solving. Engineering is not about guessing. When something does not work, it is important to test carefully. In digital design, I used simulation tools to observe signals and waveforms. In embedded systems, I tested small parts of code before running the full program. I learned to follow a clear process: understand the problem, think of possible causes, test one solution at a time, and confirm the final result. This structured method helps reduce mistakes and saves time. I also learned to think about the impact of engineering solutions. In society, technology affects people's safety and privacy. For example, wireless devices collect data, and if security is weak, users can be harmed. Engineers must design systems carefully and responsibly. In the economic context, good

design can reduce cost and improve efficiency. Simple and reliable systems save money. In the environmental context, power efficiency is important. Low-power devices reduce energy use and electronic waste. In the global context, devices must follow international standards so they can work worldwide. These lessons helped me understand that engineering is connected to real life. Using outside resources was also an important part of my growth. Sometimes classroom lessons were not enough to solve real problems. I learned to read documentation, search for tutorials, and study extra materials. I read about debugging methods, bus interfaces, memory systems, and Bluetooth architecture. I realized that learning from outside sources is part of being an engineer. Real engineers always read manuals and continue learning new tools.

Finally, I understand that engineering requires life-long learning. Technology changes quickly, and new systems are developed every year. My short term goal is to strengthen my knowledge in digital design, verification, and embedded systems. My long-term goal is to work in hardware and chip design. To achieve this, I plan to practice regularly, build small projects, and set weekly learning goals. In conclusion, my college experience helped me grow in communication, technical knowledge, and problem-solving. My classes gave me strong foundations and real project experience. I learned to break down complex systems, debug step by step, and think about the impact of engineering in the real world. Most importantly, I learned that becoming a strong engineer is a continuous journey of learning and improvement.